

Incident report — MTAPI bridge Docker logs have no size limit

Unbounded container log growth on the production VPS · 7 October 2026

Severity	Low — disk is 7% full, but one log already holds 2.9 GB and nothing stops it growing	Status	OPEN diagnosed, not yet fixed
Detected	7 October 2026	Component	VPS Docker logging (<code>mtapi</code> , <code>mt4rest</code>)
Affected users	All — the bridge serves every trade on the platform	Root cause	Containers created with no log driver and no log options → Docker's unbounded <code>json-file</code> default
Reported by	Found during the "Single target TP" investigation	Data impact	None — no customer data at risk

1. Executive summary (plain English)

The MTAPI bridge — the small service on our VPS that actually places and modifies trades on the MetaTrader terminal — writes a log line for **every request it receives**, and each line contains the full request URL, which is long. Docker's default behaviour is to keep every line forever in a single file, with **no maximum size and no rotation**.

Nothing in our setup caps that file. Our container installation script does not pass a log size limit, we have no Docker daemon configuration file that sets a default, and the only time we ever touch a log file is to wipe it during a security reinstall.

Measured state today: the disk is healthy — only **7% used (6.5 GB of 96 GB)**. But one container's log file has already grown to **2.9 GB**, roughly a third of everything stored on the machine, and it is still growing. Nothing is broken yet. The risk is silent growth until the disk fills, at which point Docker and the trading terminal start failing in ways that are painful to diagnose.

Why this matters more than usual: this log is **the only place** we can see exactly what the broker was asked to do. It was essential evidence in today's trading investigation, so losing it — or having it roll over — directly costs us debugging capability.

The fix is small: give the container a log size limit when we create it.

2. Issue encountered

Investigating an unrelated trade problem, we needed the bridge's own request log to see what the broker had actually been asked to do. That log held the only copy of the answer. It then raised a second question: how far back does this log go, and how big is it?

a) Measurements taken on the production VPS (2026-10-07)

Overall disk

```
$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        96G   6.5G   90G   7% /
```

Container log files (the unbounded ones)

```
$ du -sh /var/lib/docker/containers/*/*-json.log | sort -h | tail -10
110M   .../ac4da626a5fd.../ac4da626a5fd...-json.log
2.9G   .../f92a9d631a24.../f92a9d631a24...-json.log
```

The 2.9 GB file is the bridge's log — the container handling every trade request. Map container IDs to names with `docker ps -a --format '{{.ID}} {{.Names}}'` .

Confirming no size limit is configured

```
$ docker ps -a --format '{{.Names}}' | xargs -I{} docker inspect -f \
'{{.Name}} driver={{.HostConfig.LogConfig.Type}} opts={{.HostConfig.LogConfig.Config}}' {}
/mt4rest driver=json-file opts=map[]
/mtapi driver=json-file opts=map[]
```

`opts=map[]` is an empty options map — proof that neither container has a size limit.

Confirming there is no daemon-wide default

```
$ cat /etc/docker/daemon.json 2>/dev/null || echo "no daemon.json"
no daemon.json
```

nginx access and error logs (separate files)

```
$ ls -lh /var/log/nginx/
total 237M
-rw-r----- 1 www-data adm    65M Oct  8 00:00 mtapi-access.log
-rw-r----- 1 www-data adm  101M Oct  7 00:00 mtapi-access.log.1
-rw-r----- 1 www-data adm   2.1M Oct  6 00:00 mtapi-access.log.2.gz
... daily .gz files back to Sep 21 ...
```

Good news here: nginx *is* rotating daily, roughly two weeks of compressed history is kept, and this is not an active problem. It is noted because incident investigations depend on it.

b) Two findings

- The Docker container log has no retention policy at all.** Docker's default `json-file` driver is in use because nothing overrides it. That driver writes to a single file per container and never rotates or truncates it on its own.
- The only log-size handling in the entire repository is a security wipe.** The installation script truncates log files to zero **only** as part of removing evidence of a previously exposed password (the 2026-09-23 incident). That is a one-off cleanup, not a size policy.

The 2026-09-23 incident report already listed *"Optionally cap Docker log size for future restarts"* as a suggested follow-up. It was never actioned.

3. Affected user(s)

Not user-specific. Every trade placed through the bridge contributes lines to this log, so growth scales with platform activity rather than with any one customer.

Item	Value
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Containers affected	mtapi (the bridge), mt4rest
Log driver in use	Docker default json-file
Options configured	None (opts=map[])
Largest log file	2.9 GB
Other container log	110 MB
Disk usage	6.5 GB of 96 GB (7%)
Rotation	None
nginx access log	65 MB current + 101 MB previous, rotating daily — working, unaffected

4. Root cause

deploy/mtapi-log-redactor/install.sh creates the container with docker run -d at line 139 and passes --name , --restart always , network/port/environment/volume arguments and a custom entrypoint — but no --log-driver and no --log-opt :

```
docker run -d \  
  --name "$CONTAINER" \  
  --restart always \  
  ...  
  --entrypoint /mtapi-redactor/entrypoint.sh \  
  "$IMAGE"
```

- Docker falls back to its compiled-in default, the json-file driver.
- The default json-file driver has no max-size and no max-file , so the file grows forever.
- There is no /etc/docker/daemon.json on the VPS (verified above), so no daemon-wide default applies either.
- install.sh lines 188 and 192 call truncate -s 0 on the container log — but only to scrub credentials during a reinstall, and only for the specific container IDs involved.

The log redactor (deploy/mtapi-log-redactor/redact.awk) strips account passwords from lines as they are written, so this is not a secret-exposure issue (that was the separate 2026-09-23 incident). It is purely an unbounded-growth issue.

5. The fix

Step A — reclaim the space (the prune)

Two options. Option 1 is preferred because it does not interrupt trading.

Option 1 — reclaim by redeploying (no trading interruption). Creating a container gives it a brand-new log file, and removing the old container removes its log directory with it, so applying the fix and rerunning the installer reclaims the space as a side effect:

```
cd /root/TSCopier # or wherever the repo lives on the VPS  
docker rm -f mtapi  
bash deploy/mtapi-log-redactor/install.sh  
du -sh /var/lib/docker/containers/*/*-json.log | sort -h | tail -10  
df -h /
```

Option 2 — reclaim immediately without redeploying (brief trading interruption). Stopping the containers pauses trade execution for a few seconds, so only do this in a quiet window — outside market hours or at the weekend.

```
docker stop mtapi mt4rest  
find /var/lib/docker/containers -name '*-json.log' -exec truncate -s 0 {} +  
docker start mtapi mt4rest  
df -h /
```

Truncating the log of a running container is not reliable, which is why Option 2 stops the containers first. Option 1 avoids the question entirely.

Step B — cap the log so it cannot happen again

1. Add log options to the container creation in deploy/mtapi-log-redactor/install.sh , inside the docker run -d block at line 139:

```
--log-driver json-file \  
--log-opt max-size=50m \  
--log-opt max-file=3 \  

```

This caps each container at 150 MB worst case (50 MB × 3 files).

2. Add a daemon-wide default at /etc/docker/daemon.json on the VPS, so containers we create later — including mt4rest , which is not created by this script — are capped too:

```
{  
  "log-driver": "json-file",  
  "log-opts": { "max-size": "50m", "max-file": "3" }  
}
```

Apply with systemctl restart docker . **Important:** these defaults only affect containers created after the restart; existing containers keep the settings they were created with, so they must still be recreated (Option 1 above).

3. nginx needs no change. Rotation was verified as working (daily .gz history back to 21 September). Keep it as-is and re-check after any nginx reconfiguration.

6. Files changed

Nothing has been changed yet. The planned changes are:

File	Change
deploy/mtapi-log-redactor/install.sh	Add --log-driver json-file --log-opt max-size=50m --log-opt max-file=3 to the docker run -d block (line 139)
VPS /etc/docker/daemon.json	Create with daemon-wide log defaults (VPS only, not in the repo)
docs/incidents/incident-2026-10-07-mtapi-bridge-log-	This report

retention.md	
docs/PROJECT_MEMORY.md	Changelog entry

7. Verification

After the fix is deployed:

- `docker inspect -f '{{.HostConfig.LogConfig}}' mtapi` must show `json-file map[max-file:3 max-size:50m]`.
- `df -h /` must show the reclaimed space (baseline: 6.5 GB used, 7%).
- `du -sh /var/lib/docker/containers/*/*-json.log | sort -h | tail -10` must show every log file under 150 MB (baseline: 2.9 GB and 110 MB).
- `cat /etc/docker/daemon.json` must exist and contain the log options.
- Trade normally for a few days and confirm log files stop growing past the cap.

8. Deployment status

Not deployed. This is a VPS-side change and is not covered by the Railway or Netlify deployments. It must be applied manually on the production VPS. Because the disk is only 7% full, this is **not urgent** — it can be done in a normal maintenance window. Recreating the container briefly interrupts trade execution, so schedule it outside market hours.

9. Follow-ups

- Apply Step A and Step B on the VPS in a quiet market window.
- Add the log cap to the MTAPI cutover runbook so a future rebuild does not reintroduce it.
- Decide whether the bridge request log should also be kept for a fixed number of days rather than only capped by size — incident investigations depend on being able to look back at it.
- Confirmed working, no action:** nginx access-log rotation (daily, ~2 weeks of history).
- Re-measure the disk after the fix and record the "after" numbers in this report.